

Recursive Asymptotic Variance Estimation in Continuous-Time Markov Chains (CTMC)

Pedagogical Peer Instruction
For Graduate-Level Stochastic Control & RL

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Abstract

This document derives a novel, recursive, online algorithm for estimating the asymptotic variance of a reward function defined on a Continuous-Time Markov Chain (CTMC) using single-trajectory samples. Our approach represents the continuous-time analog of the linear stochastic approximation (SA) method proposed by Agrawal, Prashanth, and Maguluri (2024) for Discrete-Time Markov Chains (DTMCs). We utilize the continuous-time Poisson equation to construct a temporal difference (TD)-like learning scheme that requires $O(1)$ computation and memory per step. By scaling the stochastic approximation updates with observed holding times, we show that the jump-chain discrete-time updates converge exactly to continuous-time stationary expectations. We provide a rigorous analysis of the value function estimator's stability and present extensions under linear function approximation (LFA).

1 Introduction and Physical Intuition

In stochastic processes, moving from discrete-time Markov chains (DTMCs) to continuous-time Markov chains (CTMCs) represents a transition from discrete steps to a continuous flow. This transition simplifies the mathematical structure of long-term fluctuations.

To build physical intuition, consider a traveler wandering through a house with several rooms. Each room $x \in S$ has a specific room temperature $f(x)$.

- **Discrete-Time (DTMC):** The traveler teleports to a new room once every minute (discrete steps).
- **Continuous-Time (CTMC):** The traveler stays in each room x for a continuous, exponentially distributed duration of time with rate $q(x)$ before moving to another room.

Over a very long journey, the traveler spends a fraction of their time in room x given by the stationary probability π_x . The long-run average temperature they experience is the **stationary mean**:

$$\bar{f} = \mathbb{E}_\pi[f(X)] = \sum_{x \in S} \pi_x f(x) \quad (1)$$

2 The Value Function and the Poisson Equation

To understand the fluctuations around the stationary mean, we define the **value function** $V(x)$ (also called the expected head-start or deviation function). This function answers the question:

“If the traveler starts in room x today, what is the net cumulative temperature difference they will experience over their entire future journey compared to someone who starts in the long-run average condition?”

Formally, the value function $V : S \rightarrow \mathbb{R}$ is defined as:

$$V(x) = \int_0^\infty \mathbb{E}[f(X_t) - \bar{f} \mid X_0 = x] dt \quad (2)$$

In a CTMC, the transition rates are governed by the infinitesimal generator matrix Q , where Q_{xy} is the transition rate from state x to state y for $x \neq y$, and $Q_{xx} = -q(x) = -\sum_{y \neq x} Q_{xy}$. By the properties of the generator, the expected rate of change of our future head-start must perfectly balance the instantaneous temperature deviation:

$$QV(x) = -(f(x) - \bar{f}) \quad (3)$$

In vector-matrix form, this is the **continuous-time Poisson equation**:

$$-Q\mathbf{V} = \mathbf{f} - \bar{f}\mathbf{1} \quad (4)$$

Proof of the Poisson Equation. By definition of the generator matrix Q acting on the function V :

$$QV(x) = \lim_{h \rightarrow 0^+} \frac{\mathbb{E}[V(X_h) \mid X_0 = x] - V(x)}{h} \quad (5)$$

Substituting the definition of $V(x)$:

$$\begin{aligned} \mathbb{E}[V(X_h) \mid X_0 = x] &= \int_0^\infty \mathbb{E}[f(X_{t+h}) - \bar{f} \mid X_0 = x] dt \\ &= \int_h^\infty \mathbb{E}[f(X_s) - \bar{f} \mid X_0 = x] ds \end{aligned} \quad (6)$$

Subtracting $V(x)$ from both sides:

$$\mathbb{E}[V(X_h) \mid X_0 = x] - V(x) = - \int_0^h \mathbb{E}[f(X_s) - \bar{f} \mid X_0 = x] ds \quad (7)$$

Dividing by h and taking the limit as $h \rightarrow 0^+$:

$$\begin{aligned} QV(x) &= \lim_{h \rightarrow 0^+} \frac{-1}{h} \int_0^h \mathbb{E}[f(X_s) - \bar{f} \mid X_0 = x] ds \\ &= -(f(x) - \bar{f}) \end{aligned} \quad (8)$$

where the last equality follows from the Fundamental Theorem of Calculus since the integrand is continuous. $\square$

3 Asymptotic Variance $\kappa(f)$

If the traveler journeys for a long duration T , the total accumulated temperature they experience is the integral:

$$I_T = \int_0^T f(X_s) ds \quad (9)$$

By the Ergodic Theorem, $I_T/T \rightarrow \bar{f}$ almost surely as $T \rightarrow \infty$. By the Central Limit Theorem (CLT) for continuous-time processes, the scaled fluctuations converge in distribution:

$$\frac{I_T - T\bar{f}}{\sqrt{T}} \Rightarrow \mathcal{N}(0, \kappa(f)) \quad \text{as } T \rightarrow \infty \quad (10)$$

The parameter $\kappa(f)$ is the **asymptotic variance** of the reward function on the CTMC:

$$\kappa(f) = \lim_{T \rightarrow \infty} \frac{1}{T} \text{Var} \left(\int_0^T f(X_s) ds \right) \quad (11)$$

Asymptotic variance is different from the stationary variance $\text{Var}_\pi(f(X)) = \mathbb{E}_\pi[(f(X) - \bar{f})^2]$. It captures the temporal correlations (or “stickiness”) of the states. Mathematically, it integrates all autocovariances over time:

$$\kappa(f) = 2 \int_0^\infty \text{Cov}_\pi(f(X_0), f(X_t)) dt \quad (12)$$

4 Derivation of $\kappa(f)$ via the Poisson Equation

Let $Z_t = f(X_t) - \bar{f} = f(X_t) - \mathbb{E}_\pi[f(X)]$ be the centered reward process. Assuming the process starts in steady state ($X_0 \sim \pi$), we have:

$$\begin{aligned} \text{Var} \left(\int_0^T Z_s ds \right) &= \mathbb{E}_\pi \left[\left(\int_0^T Z_s ds \right)^2 \right] \\ &= \int_0^T \int_0^T \mathbb{E}_\pi[Z_u Z_v] du dv \\ &= 2 \int_0^T \int_0^v \mathbb{E}_\pi[Z_u Z_v] du dv \end{aligned} \quad (13)$$

By stationarity, $\mathbb{E}_\pi[Z_u Z_v] = \mathbb{E}_\pi[Z_0 Z_{|v-u|}]$. Changing variables using $\tau = v - u$:

$$\text{Var} \left(\int_0^T Z_s ds \right) = 2 \int_0^T (T - \tau) \mathbb{E}_\pi[Z_0 Z_\tau] d\tau \quad (14)$$

Dividing by T and taking the limit as $T \rightarrow \infty$ gives the asymptotic variance:

$$\begin{aligned} \kappa(f) &= \lim_{T \rightarrow \infty} \frac{2}{T} \int_0^T (T - \tau) \mathbb{E}_\pi[Z_0 Z_\tau] d\tau \\ &= 2 \int_0^\infty \mathbb{E}_\pi[Z_0 Z_\tau] d\tau \end{aligned} \quad (15)$$

Using the law of total expectation, we can express the integrand as:

$$\mathbb{E}_\pi[Z_0 Z_\tau] = \mathbb{E}_\pi[Z_0 \mathbb{E}[Z_\tau | X_0]] \quad (16)$$

Substituting this back into our expression for $\kappa(f)$:

$$\begin{aligned} \kappa(f) &= 2 \int_0^\infty \mathbb{E}_\pi [(f(X_0) - \bar{f}) \mathbb{E}[f(X_\tau) - \bar{f} | X_0]] d\tau \\ &= 2 \mathbb{E}_\pi \left[(f(X_0) - \bar{f}) \int_0^\infty \mathbb{E}[f(X_\tau) - \bar{f} | X_0] d\tau \right] \end{aligned} \quad (17)$$

Recognizing the inner integral as the value function $V(X_0)$, we obtain the elegant formula:

$$\kappa(f) = 2 \mathbb{E}_\pi [(f(X) - \bar{f}) V(X)] \quad (18)$$

5 The Elegant Contrast: DTMC vs. CTMC

To appreciate the simplicity of continuous time, let us compare the asymptotic variance formulas side-by-side:

$$\kappa_{\text{DTMC}}(f) = 2\mathbb{E}_\pi[(f(X) - \bar{f})V_{\text{DTMC}}(X)] - \mathbb{E}_\pi[(f(X) - \bar{f})^2] \quad (19)$$

$$\kappa_{\text{CTMC}}(f) = 2\mathbb{E}_\pi[(f(X) - \bar{f})V(X)] \quad (20)$$

Why does the negative term disappear in CTMC?

In discrete time, when we calculate the variance of a sum of random variables $S_n = \sum_{k=0}^{n-1} Z_k$, the diagonal covariance terms $\text{Cov}(Z_k, Z_k)$ are isolated, discrete point masses. Under steady state, each contributes $\mathbb{E}_\pi[Z_0^2]$. In the double summation, these diagonal terms are counted once, while the off-diagonal terms $\text{Cov}(Z_i, Z_j)$ are counted twice. To relate the total sum to $2 \sum_{k=0}^{\infty} \mathbb{E}_\pi[Z_0 Z_k]$, we must subtract the single-step variance $\mathbb{E}_\pi[Z_0^2]$ to correct for the double-counting of the diagonal.

In continuous time, there is no discrete “diagonal.” The double integral integrates over a continuous 2D domain $[0, T] \times [0, T]$. The diagonal line $u = v$ has a Lebesgue measure of zero in $\mathbb{R}^2$. Therefore, the instantaneous variance at the points $u = v$ contributes nothing to the continuous integral!

As a result, the continuous-time asymptotic variance is mathematically cleaner, more symmetric, and does not require any discrete correction terms.

6 Stochastic Approximation in CTMC: Tabular Setting

Our goal is to estimate $\kappa(f)$ recursively using a single trajectory of a CTMC. A continuous-time trajectory is naturally observed at transition times:

$$(X_0, \tau_0), (X_1, \tau_1), \dots, (X_k, \tau_k) \quad (21)$$

where X_k is the state occupied during the k -th interval $[T_k, T_{k+1})$, and $\tau_k = T_{k+1} - T_k > 0$ is the continuous holding time spent in state X_k before jumping to state X_{k+1} . Under the CTMC dynamics, τ_k is exponentially distributed with rate $q(X_k)$, so $\mathbb{E}[\tau_k | X_k] = 1/q(X_k)$.

6.1 Weighted Average Reward Estimation

The stationary expectation $\bar{f} = \mathbb{E}_\pi[f(X)]$ is the time-average reward. Since the time spent in state X_k is τ_k , we can construct a recursive estimator $\bar{f}_k$ as follows:

$$\bar{f}_{k+1} = \bar{f}_k + c_1 \alpha_k \tau_k (f(X_k) - \bar{f}_k) \quad (22)$$

Let us verify the convergence of this estimator. Under the stationary distribution of the jump chain π^J , the expected update step is:

$$\mathbb{E}_{\pi^J}[\tau_k (f(X_k) - \bar{f}_k)] = \mathbb{E}_{\pi^J}[\mathbb{E}[\tau_k | X_k] (f(X_k) - \bar{f}_k)] = \sum_{x \in S} \pi_x^J \frac{f(x) - \bar{f}_k}{q(x)} \quad (23)$$

Since the stationary distribution of the continuous chain π relates to the jump-chain stationary distribution π^J via $\pi_x = \frac{\pi_x^J / q(x)}{\sum_y \pi_y^J / q(y)}$, we have:

$$\sum_{x \in S} \pi_x^J \frac{f(x) - \bar{f}_k}{q(x)} = 0 \iff \bar{f}_k \rightarrow \frac{\sum_x \pi_x^J f(x) / q(x)}{\sum_x \pi_x^J / q(x)} = \sum_{x \in S} \pi_x f(x) = \bar{f} \quad (24)$$

Thus, multiplying the update by the holding time τ_k acts as the mathematically exact importance-sampling weight to reconcile the jump-chain state distribution with the continuous-time stationary distribution π .

6.2 Value Function and Temporal Difference Learning

Dividing the continuous-time Poisson equation $QV(x) = -(f(x) - \bar{f})$ by $q(x)$ yields:

$$V(x) - \sum_{y \neq x} \frac{q(x, y)}{q(x)} V(y) = \frac{f(x) - \bar{f}}{q(x)} \quad (25)$$

Recognizing $\sum_{y \neq x} \frac{q(x, y)}{q(x)} V(y) = \mathbb{E}[V(X_{k+1}) \mid X_k = x]$ under the jump chain, we define the **Continuous-Time Temporal Difference (TD) error**:

$$\delta_k = \tau_k(f(X_k) - \bar{f}_k) + V_k(X_{k+1}) - V_k(X_k) \quad (26)$$

Indeed, its conditional expectation given X_k and V_k is:

$$\begin{aligned} \mathbb{E}[\delta_k \mid X_k = x, V_k] &= \mathbb{E}[\tau_k \mid X_k = x](f(x) - \bar{f}_k) + \mathbb{E}[V_k(X_{k+1}) \mid X_k = x] - V_k(x) \\ &= \frac{1}{q(x)}(f(x) - \bar{f}_k) + \sum_{y \neq x} \frac{q(x, y)}{q(x)} V_k(y) - V_k(x) \\ &= \frac{1}{q(x)} [(QV_k)(x) + f(x) - \bar{f}_k] \end{aligned} \quad (27)$$

When $V_k = V$ and $\bar{f}_k = \bar{f}$, this expectation is precisely zero, fulfilling the Poisson equation.

To estimate V , we perform TD updates with the same projection onto the subspace orthogonal to $\mathbf{1}$ used in the discrete-time paper:

$$V_{k+1}(x) = \begin{cases} V_k(x) + \alpha_k \delta_k (1 - 1/S), & \text{for } x = X_k, \\ V_k(x) - \alpha_k \delta_k / S, & \text{otherwise.} \end{cases} \quad (28)$$

6.3 Asymptotic Variance Estimation

Using Equation (20), we expand the asymptotic variance:

$$\kappa(f) = 2\mathbb{E}_\pi[(f(X) - \bar{f})V(X)] = 2\mathbb{E}_\pi[f(X)V(X)] - 2\bar{f}\bar{V} \quad (29)$$

where $\bar{V} = \mathbb{E}_\pi[V(X)]$. This is mathematically equivalent to:

$$\kappa(f) = \mathbb{E}_\pi[2f(X)V(X) - 2f(X)\bar{V}] \quad (30)$$

To estimate $\bar{V}$ and $\kappa(f)$ online, we use holding-time weighted stochastic approximation:

$$\bar{V}_{k+1} = \bar{V}_k + c_2 \alpha_k \tau_k (V_k(X_k) - \bar{V}_k) \quad (31)$$

$$\kappa_{k+1} = (1 - c_3 \alpha_k \tau_k) \kappa_k + c_3 \alpha_k \tau_k (2f(X_k)V_k(X_k) - 2f(X_k)\bar{V}_k) \quad (32)$$

Algorithm 1:	CTMC Asymptotic Variance Estimator (Tabular Setting)
Input:	Time horizon $n > 0$, constants $c_1, c_2, c_3 > 0$, step-size sequence $\{\alpha_k\}$.
Initialization:	$\bar{f}_0 = 0, V_0 = \mathbf{0}, \bar{V}_0 = 0$, and $\kappa_0 = 0$.
Loop:	for $k \leftarrow 0$ to $n - 1$ do Observe $(X_k, f(X_k), \tau_k, X_{k+1})$. // 1. Average reward $\bar{f}$ estimation $\bar{f}_{k+1} = \bar{f}_k + c_1 \alpha_k \tau_k (f(X_k) - \bar{f}_k)$ // 2. Temporal Difference (TD) error $\delta_k = \tau_k (f(X_k) - \bar{f}_k) + V_k(X_{k+1}) - V_k(X_k)$ // 3. Value function V estimation with projection $V_{k+1}(x) = \begin{cases} V_k(x) + \alpha_k \delta_k (1 - 1/S), & \text{if } x = X_k \\ V_k(x) - \alpha_k \delta_k / S, & \text{otherwise} \end{cases}$ // 4. Average value function $\bar{V}$ estimation $\bar{V}_{k+1} = \bar{V}_k + c_2 \alpha_k \tau_k (V_k(X_k) - \bar{V}_k)$ // 5. Asymptotic Variance $\kappa(f)$ estimation $\kappa_{k+1} = (1 - c_3 \alpha_k \tau_k) \kappa_k + c_3 \alpha_k \tau_k (2f(X_k)V_k(X_k) - 2f(X_k)\bar{V}_k)$
Output:	Estimates $\kappa_n, \bar{V}_n, V_n$, and $\bar{f}_n$.

7 Convergence Analysis and Stability

We can express the tabular V update as a linear stochastic approximation algorithm. In vector form, let $\mathbf{V}_k \in \mathbb{R}^S$ denote the value function estimate, and let $\Pi = I - \frac{1}{S} \mathbf{1}\mathbf{1}^T$ be the projection matrix. The update is:

$$\mathbf{V}_{k+1} = \mathbf{V}_k + \alpha_k \Pi (\tau_k (\mathbf{f} - \bar{f}_k \mathbf{1}) e_{X_k} + (e_{X_{k+1}} - e_{X_k}) \mathbf{V}_k(X_k)) \quad (33)$$

The expected direction of the update at state $X_k = x$, given the current value function V , is:

$$\begin{aligned}
\bar{g}(\mathbf{V}) &= \mathbb{E}_{\pi^J} [\mathbb{E}[\Pi \delta_k e_{X_k} \mid X_k]] \\
&= \Pi \sum_{x \in S} \pi_x^J \frac{1}{q(x)} [(Q\mathbf{V})(x) + f(x) - \bar{f}] e_x \\
&= \Pi D_{\pi^J} D_q^{-1} (Q\mathbf{V} + \mathbf{f} - \bar{f} \mathbf{1})
\end{aligned} \quad (34)$$

where $D_{\pi^J} = \text{diag}(\pi_1^J, \dots, \pi_S^J)$ and $D_q = \text{diag}(q(1), \dots, q(S))$. Recall that $\pi_x = \frac{\pi_x^J / q(x)}{\sum_y \pi_y^J / q(y)}$, which implies $D_{\pi^J} D_q^{-1} \propto D_\pi = \text{diag}(\pi_1, \dots, \pi_S)$. Thus, the expected average update matrix is:

$$A = \Pi D_\pi Q \quad (35)$$

Since we restrict our search space to $\mathcal{E} = \{\mathbf{V} \in \mathbb{R}^S : \mathbf{1}^T \mathbf{V} = 0\}$, the projection matrix acts as the identity operator on $\mathcal{E}$. To establish the stability of our linear stochastic approximation, we need to show that the matrix A is Hurwitz on the subspace $\mathcal{E}$ (i.e., all its eigenvalues restricted to $\mathcal{E}$ have strictly negative real parts).

Theorem 1. *Let Q be the infinitesimal generator of an irreducible and positive recurrent CTMC, and let D_π be the diagonal matrix of its stationary distribution. Then the matrix $A = D_\pi Q$ is Hurwitz on the subspace $\mathcal{E} = \{\mathbf{V} \in \mathbb{R}^S : \mathbf{1}^T \mathbf{V} = 0\}$. That is,*

$$\mathbf{V}^T (D_\pi Q) \mathbf{V} < 0 \quad \text{for all non-zero } \mathbf{V} \in \mathcal{E} \quad (36)$$

Proof. Let $P_t = e^{Qt}$ be the transition probability semigroup of the CTMC. For any $t > 0$, P_t is a transition probability matrix of an irreducible and aperiodic DTMC with the same stationary distribution π . By Tsitsiklis and Van Roy (1997, Lemma 7), we have:

$$\mathbf{V}^T D_\pi (I - P_t) \mathbf{V} > 0 \quad \text{for all non-zero } \mathbf{V} \in \mathcal{E} \quad (37)$$

Expanding $P_t = I + Qt + o(t)$ for small $t > 0$, we get:

$$\mathbf{V}^T D_\pi (I - I - Qt - o(t)) \mathbf{V} > 0 \implies -t \mathbf{V}^T D_\pi Q \mathbf{V} - o(t) > 0 \quad (38)$$

Dividing by $t > 0$ and taking the limit as $t \rightarrow 0^+$:

$$-\mathbf{V}^T D_\pi Q \mathbf{V} \geq 0 \quad (39)$$

To show strictly greater than, suppose there exists a non-zero $\mathbf{V} \in \mathcal{E}$ such that $\mathbf{V}^T D_\pi Q \mathbf{V} = 0$. Since $-Q$ is a positive semidefinite operator with null space spanned only by $\mathbf{1}$, and D_π is positive definite, $\mathbf{V}^T D_\pi Q \mathbf{V} = 0$ if and only if $\mathbf{V}$ is a constant vector (i.e., $\mathbf{V} \propto \mathbf{1}$). However, since $\mathbf{V} \in \mathcal{E}$, we have $\mathbf{1}^T \mathbf{V} = 0$. The only vector that is both a constant vector and orthogonal to $\mathbf{1}$ is the zero vector, which contradicts the assumption that $\mathbf{V}$ is non-zero. Hence,

$$\mathbf{V}^T (D_\pi Q) \mathbf{V} < 0 \quad \text{for all non-zero } \mathbf{V} \in \mathcal{E} \quad (40)$$

This contractive property guarantees that the average update matrix is negative definite on the subspace $\mathcal{E}$, implying that all eigenvalues of A restricted to $\mathcal{E}$ have strictly negative real parts (Hurwitz), ensuring robust, asymptotically stable convergence of the SA algorithm. $\square$

8 Linear Function Approximation Setting (LFA)

In settings where the state space S is very large, the tabular representation of V is computationally intractable. We approximate the value function by projecting it onto a lower-dimensional linear subspace spanned by a set of d linearly independent feature vectors:

$$V(s) \approx \phi(s)^T \theta \quad (41)$$

where $\phi(s) \in \mathbb{R}^d$ is the feature vector for state s , and $\theta \in \mathbb{R}^d$ is the parameter vector.

8.1 TD Updates in LFA

The Continuous-Time Temporal Difference error under LFA is defined as:

$$\delta_k = \tau_k (f(X_k) - \bar{f}_k) + (\phi(X_{k+1}) - \phi(X_k))^T \theta_k \quad (42)$$

We project the parameter updates onto the subspace E orthogonal to the constant feature direction (if it exists) using the projection matrix $\Pi_{2,E}$:

$$\theta_{k+1} = \theta_k + \alpha_k \Pi_{2,E} \phi(X_k) \delta_k \quad (43)$$

The auxiliary parameters for the average value function and the asymptotic variance are updated using holding-time weights:

$$\tilde{V}_{k+1} = \tilde{V}_k + c_2 \alpha_k \tau_k (\phi(X_k)^T \theta_k - \tilde{V}_k) \quad (44)$$

$$\kappa_{k+1} = (1 - c_3 \alpha_k \tau_k) \kappa_k + c_3 \alpha_k \tau_k \left(2f(X_k) \phi(X_k)^T \theta_k - 2f(X_k) \tilde{V}_k \right) \quad (45)$$

Algorithm 2:	CTMC Asymptotic Variance Estimator with LFA Setting
Input:	Time horizon $n > 0$, constants $c_1, c_2, c_3 > 0$, step-size sequence $\{\alpha_k\}$.
Initialization:	$\bar{f}_0 = 0$, $\theta_0 = \mathbf{0}$, $\tilde{V}_0 = 0$, and $\kappa_0 = 0$.
Loop:	for $k \leftarrow 0$ to $n - 1$ do Observe $(X_k, f(X_k), \tau_k, X_{k+1})$. // 1. Average reward $\bar{f}$ estimation $\bar{f}_{k+1} = \bar{f}_k + c_1 \alpha_k \tau_k (f(X_k) - \bar{f}_k)$ // 2. Temporal Difference (TD) error $\delta_k = \tau_k (f(X_k) - \bar{f}_k) + (\phi(X_{k+1}) - \phi(X_k))^T \theta_k$ // 3. Feature weight θ estimation with projection $\theta_{k+1} = \theta_k + \alpha_k \Pi_{2,E} \phi(X_k) \delta_k$ // 4. Average value function $\tilde{V}$ estimation $\tilde{V}_{k+1} = \tilde{V}_k + c_2 \alpha_k \tau_k (\phi(X_k)^T \theta_k - \tilde{V}_k)$ // 5. Asymptotic Variance $\kappa(f)$ estimation $\kappa_{k+1} = (1 - c_3 \alpha_k \tau_k) \kappa_k + c_3 \alpha_k \tau_k \left(2f(X_k) \phi(X_k)^T \theta_k - 2f(X_k) \tilde{V}_k \right)$
Output:	Estimates κ_n , $\tilde{V}_n$, θ_n , and $\bar{f}_n$.

9 Conclusion

We have derived a fully recursive, online, $O(1)$ -computation and memory algorithm for estimating the asymptotic variance of a function defined on an irreducible and positive recurrent CTMC. Unlike its DTMC counterpart, the continuous-time estimator is mathematically cleaner and possesses no discrete point-mass variance offset terms, thanks to the continuous domain of integration. By scaling each stochastic approximation step with the state holding times, the estimator autonomously accounts for CTMC state-transition times, achieving robust convergence to the true continuous-time variance. This algorithm can be deployed as a critical component in risk-sensitive continuous-time reinforcement learning and continuous MCMC diagnostic tracking.